

Computational Approach to Neorealism and Constructivism

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January 2026
Draft Proposal
Version 1

Abstract

Thirty years since A. Wendt's Constructivism challenged K. Waltz's Neorealism both International Relations (IR) theories are acknowledged as major pillars to the field. Although Agent-based modelling (ABM) has long made its way into IR, direct comparisons in the same simulation between neorealism and constructivism have been absent. Recent advancements in Artificial Intelligence allow for a comparative study as constructivist actor macro-behaviour can be modelled through Large-Language Model (LLM) agents. Reinforcement Learning (RL) agents represent materialistic Neorealist actors.

My thesis will explore under what material conditions the representations of Neorealist actors and Constructivist actors will produce divergent emergent social structures.

1. Research Context

The ending of the Cold War introduced an intellectual vacuum as existing paradigms were unable to predict its peaceful conclusion. **Neorealism**, as outlined by Kenneth Waltz [5], theorizes that states act out of self-interest, be it for survival or power accumulation, given the anarchical system that is the world stage. The USSR undermining their relative gain to the US could be explained through the lens of upcoming **Constructivism**, formalized by Alexander Wendt [6] in 1992 as a direct criticism on Neorealism. Instead of being rigidly self-interested through material constraints, state identities and interests are formed through interaction and are changeable. As of today, both theories are recognized to have explanatory abilities.

The utilization of ABM within the IR field has extensive applications [1]. More recently both the use of RL agents [4] and LLM agents [2] has been explored. However, no direct comparison between Neorealism and Constructivism has been done and computational mapping both framework to respectively RL agents and LLM agents is novel.

Research Question: *Under what material constraints do semantic reasoning (LLM agents) and pure optimization (RL agents) produce divergent emergent social structures in an anarchic system?*

2. Research Methods

To answer the

2.1. Method Justification

General Sufficiency principle - ¿ as minimal as possible. How do we produce, with as little elements as possible, the intended macro-behaviour that we want to investigate -¿ cooperation, coalition, power hierarchies, How do I make sure it is not void of meaning. Merely driven by my intuitions as someone that doesnt even study IR?

2.1.1 Computational Mapping

RL is a widely used technique for modelling IR theories, specifically useful for materialist views such as neorealism (as we can optimize our own resources). Why?

CONSTRUCTIVISM Core thesis - identity formed through interaction, norms and group identity - interests flow from identity

LLM justification - LLMs have shown to adhere to norms and group identity through interaction (norms paper,) - LLMs show different behaviour through contextual framing (personas, language!)

2.2. Computational Model Design

The **State Space** consists of an array containing the resources of every agent and the history of all recent interactions, conflicts and their outcomes. Each round, all agents choose simultaneously from the **Action Space**:

Conflict Resolution: The resources of both sides, consisting of the main combatants plus active Arms multipliers, are summed and relative to the total resources involved, win probabilities are determined. The winner receives $T\%$ of the losers resources. All combatants pay C conflict cost.

Action	Target	Effect
Invest	Self	Costs I , generates V (net positive growth)
Invest	Other	Costs I , other receives V (signals trust)
Arm	Self	Costs D , multiplier M/O for 1 round (general threat)
Arm	Other	Costs D , multiplier M/O for 2 rounds (signals alliance)
Attack	Other	Probabilistic conflict resolution (see below)

2.3. Agent Architectures

The simulation is done using homogenous agent architectures for every agent.

Reinforcement Learning (RL) Agents represent state actors in Waltz’s theory of Neorealism. At it’s core, the agents should be self-interested and concerned with optimizing their relative gain within the system. Computationally, this identity is translated through optimizing the following function during training: [insert RL optimization function]

Large-Language Model (LLM) Agents represent state actors in Wendt’s theory of Constructivism. Based on the intersubjective interpretation of shared history, LLM agents are prompted internally update their identities and interests, and choose their next action accordingly.

Control LLM Agents represent the state actors in Neorealist theory and are prompted to optimize their relative gain through their actions. These agents are specifically introduced to effectively isolate the effects of the constructivist prompting.

2.4. Measurements

To clearly analyse the emergent social structures, three specific elements are analysed, each with specific metrics to track.

For **Coalition**:

- Coalition stability: average duration of Arm-other relationships
- Trade stability: average duration of Invest-other relationships
- Reciprocity rate: $P(A \text{ arms } B \text{ — } B \text{ armed } A \text{ last round})$
- Betrayal rate: $P(\text{attack — previous ally})$

For **Hierarchy**:

- Gini coefficient of resources
- Network centrality of Arm-other relationships

For **Cooperation/Conflict**:

- Attack frequency per round
- Invest frequency per round
- Total welfare

2.5. Validation

The use of Generative Agents within ABMs has been criticed for enlarging problems around validation and calibration that are known in ABMs. [3]. The thesis should make clear what is validated, and, importantly, what is not to prevent it being "void of meaning".

3. Significance of Research

With the debate between Neorealism and Constructivism still actively ongoing, my research could potentially provide additional insights. Though, ABM models are widely used in IR studies, the Artificial Intelligence approach to designing the agent architecture seems quite novel. Especially, comparative studies between RL and LLM behaviour in social studies is an active field of research.

References

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